System Advisor Model Report

Detailed Photovoltaic	314 DC kW Nameplate	38.89, -77.02 UTC -5
Commercial	$1.86/W Installed Cost	NSRDB

Performance Model

Modules	
Aptos Solar Technology LLC DNA-144-BF10-530W	
Cell material	Mono-c-Si Bifacial
Module area	2.58 m²
Module capacity	530.75 DC Watts
Quantity	591
Total capacity	313.67 DC kW
Total area	1,524 m²

Inverters	
SMA America: STPS27US-20	
Unit capacity	27 AC kW
MPPT Voltage Range	200 - 980 VDC
Nominal AC Voltage	208.0 VAC
Quantity	10
Total capacity	270 AC kW
DC to AC Capacity Ratio	1.16
AC losses (%)	0.00

Two subarrays:	1	2
Strings	23	21
Modules per string	12	15
String Voc (DC V)	590.40	738.00
Tilt (deg from horizontal)	20.00	20.00
Azimuth (deg E of N)	180	0
GCR	0.3	0.3
Tracking	no	no
Backtracking	-	-
Self shading	yes	yes
Rotation limit (deg)	-	-
Shading	no	no
Snow	no	no
Soiling	yes	yes
DC losses (%)	4.44	4.44

Performance Adjustments	
DC avail./curtail.	2%
AC avail./curtail.	2%
Degradation	0.5 %/yr
Hourly or custom losses	none

Annual Results (in Year 1)		
GHI kWh/m²/day	4.28	4.28
POA kWh/m²/day	110.00	85.00
Net to inverter	404,909 DC kWh	
Net to grid	372,985 AC kWh	
Capacity factor	13.6	
Performance ratio	0.8	

Financial Model

Project Costs	
Total installed cost	$583,665
Salvage value	$0

Analysis Parameters	
Project life	25 years
Inflation rate	2.5%
Real discount rate	5%

Project Debt Parameters	
Debt fraction	0%
Amount	$0
Term	25 years
Rate	7%

Tax and Insurance Rates	
Federal income tax	0 %/year
State income tax	0 %/year
Sales tax (% of indirect cost basis)	13%
Insurance (% of installed cost)	0 %/year
Property tax (% of assessed val.)	0 %/year

Incentives	
Federal ITC	30%

Electricity Usage and Rate Summary

Annual peak demand 597.8 kW
Annual total usage 1,583,134 kWh
Generic Commercial
Fixed charge: $125.96/month
Monthly excess with kWh rollover
Flat energy buy rate: $0.145/kWh

Results	
Nominal LCOE	14.1 cents/kWh
Net present value	$234,700
Payback period	7.4 years

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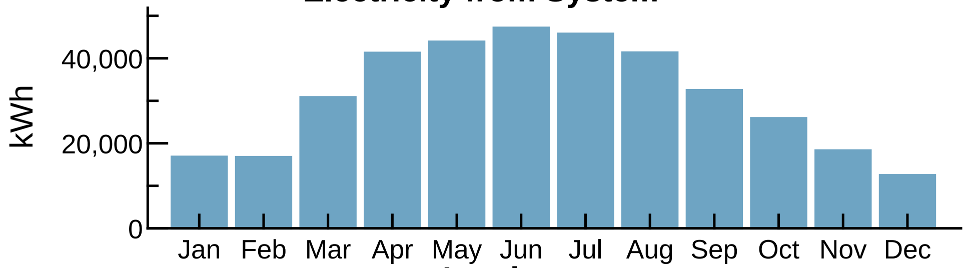
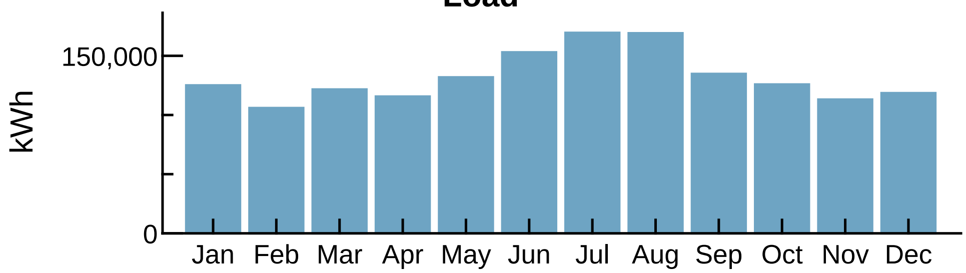
Year 1 Monthly Generation and Load Summary

Electricity from System



Load



Year 1 Monthly Electric Bill and Savings ($)

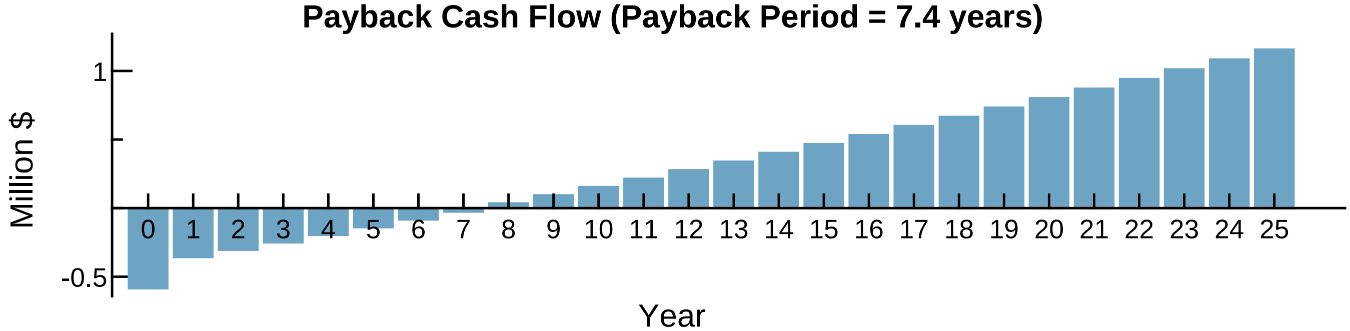
Month	Without System	With System	Savings
Jan	24,226	21,789	2,436
Feb	19,890	17,465	2,425
Mar	22,128	17,046	5,082
Apr	21,758	14,760	6,997
May	24,286	17,310	6,976
Jun	28,322	20,417	7,904
Jul	31,672	24,203	7,468
Aug	31,269	25,166	6,103
Sep	25,432	20,276	5,156
Oct	23,076	19,103	3,972
Nov	20,491	17,626	2,865
Dec	21,988	20,181	1,807
Annual	294,543	235,348	59,195

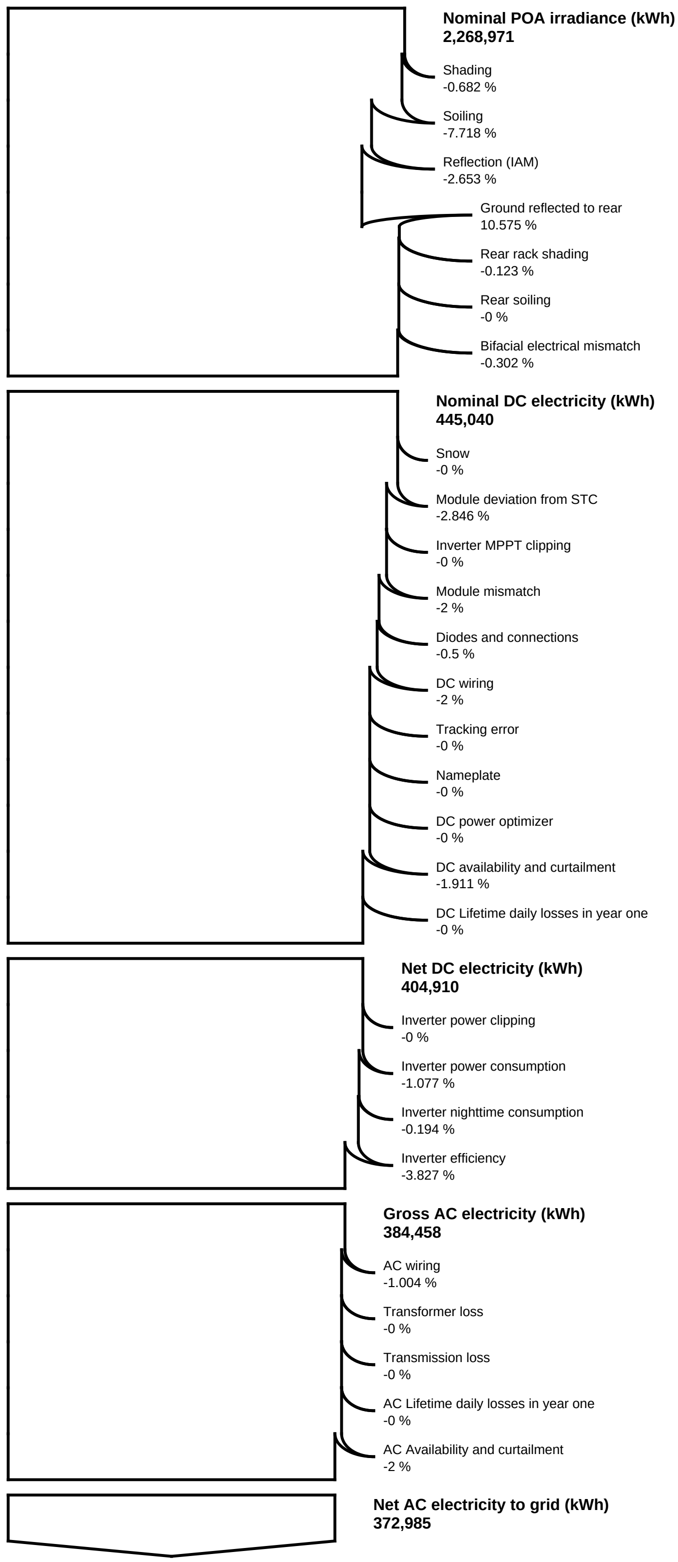
NPV Approximation using Annuities

Annuities, Capital Recovery Factor (CRF) = 0.0907		
Investment	$-52,900	Sum:
Expenses	$-10,900	$21,200
Savings	$14,700	NPV = Sum / CRF:
Energy value	$70,400	$234,000

Investment = Installed Cost - Debt Principal - IBI - CBI
Expenses = Operating Costs + Debt Payments
Savings = Tax Deductions + PBI
Energy value = Tax Adjusted Net Savings
Nominal discount rate = 7.625%

Payback Cash Flow (Payback Period = 7.4 years)




Nominal POA irradiance (kWh)
2,268,971
Shading
-0.682 %
Soiling
-7.718 %
Reflection (IAM)
-2.653 %
Ground reflected to rear
10.575 %
Rear rack shading
-0.123 %
Rear soiling
-0 %
Bifacial electrical mismatch
-0.302 %
Nominal DC electricity (kWh)
445,040
Snow
-0 %
Module deviation from STC
-2.846 %
Inverter MPPT clipping
-0 %
Module mismatch
-2 %
Diodes and connections
-0.5 %
DC wiring
-2 %
Tracking error
-0 %
Nameplate
-0 %
DC power optimizer
-0 %
DC availability and curtailment
-1.911 %
DC Lifetime daily losses in year one
-0 %
Net DC electricity (kWh)
404,910
Inverter power clipping
-0 %
Inverter power consumption
-1.077 %
Inverter nighttime consumption
-0.194 %
Inverter efficiency
-3.827 %
Gross AC electricity (kWh)
384,458
AC wiring
-1.004 %
Transformer loss
-0 %
Transmission loss
-0 %
AC Lifetime daily losses in year one
-0 %
AC Availability and curtailment
-2 %
Net AC electricity to grid (kWh)
372,985